

SANTOSH PANT

Hillsboro, OR | spant@knox.edu | [linkedin.com/in/the-santosh-pant](https://www.linkedin.com/in/the-santosh-pant) | github.com/Santoshpant23 | (309) 297-6270

EDUCATION

Knox College

Sept 2023 - June 2027

Bachelor of Science in Computer Science and Data Science (GPA: 3.82)

Galesburg, Illinois

- **Hackathon Wins:** HackIllinois 2026 (UIUC) - Best Use of Action VectorAI DB (1st, sponsor track); UMich AI for Heat Resilience - 2nd Place (\$2K); ASA DataFest 2026 - Best Insights Track; HackAugie 2026 - Best Gaming Hack
- **Honors:** Break Through Tech AI Fellow (Cornell Tech, 2025-26); Richter Research Scholar; Dean's List (6/6 terms); ICPC North Central Participant (UIUC site, 2025); Teaching Assistant - Data Structures & Algorithms; CS Club Lead

EXPERIENCE

Machine Learning Engineer - Research Collaborator

June 2026 - Present

University of Michigan - Center for Global Health Equity (Mentor: Prof. Geoffrey Siwo)

Remote

- Predicting street-level urban heat maps from RGB drone imagery; built the training and evaluation pipeline and ensembled CNN, GAN, and physically-structured models to reach 19.3 dB PSNR / 0.71 SSIM on a 202-image held-out test set (constant-prediction baseline: 10.7 dB). Code: github.com/Santoshpant23/rgb-to-thermal
- Diagnosed and fixed two silent data faults that capped accuracy (false-color palette values used as targets instead of radiometric temperature; unaligned RGB/thermal pairs from mismatched camera fields of view).

Software Engineering Intern - The MACRO Consortium

March 2025 - June 2025

- Designed and shipped a priority-based scheduling algorithm into the open-source pyscope package; resolved resource conflicts and improved allocation fairness for faculty observation requests in high-demand slots.
- Investigated and mitigated a data-loss bug in the scheduler's request-handling logic, materially improving reliability for nightly observation batches used at Knox College and partner institutions; wrote a pytest suite and refactored legacy parsing code to prevent regressions.

Software Developer Intern - Synapse Technologies

June 2024 - August 2024

- Built an internal full-stack stock market simulation platform (React, Express.js, PostgreSQL) for backtesting and evaluating quantitative trading strategies.
- Designed schemas and APIs for real-time data ingestion and P&L visualization across 50+ simulated strategy instances; improved backend performance with Redis caching and validated load characteristics with K6.

PROJECTS

Vigilant AI | FastAPI, Modal A100, YOLO11, CLIP ViT-L/14, Qwen2.5-VL, React

HackIllinois 2026 - 1st (sponsor track)

- Built an AI system in 48 hours (two-person team) that turns raw CCTV footage into structured, searchable intelligence with no manual review; won its sponsor track at HackIllinois 2026 (github.com/Santoshpant23/vigilant-ai).
- Ran object detection (YOLO11), CLIP embedding search, and a vision-language model (Qwen2.5-VL) in parallel on a serverless A100; delivered hybrid CLIP + SQLite semantic search, plain-English chat over footage with SSE streaming, and one-click PDF incident export.

Yaar | React, TypeScript, Node.js, Express, MongoDB, Gemini Live, Stripe

github.com/Santoshpant23/yaar

- Built a full-stack AI study-abroad counselor with real-time Gemini Live voice sessions over WebSocket, a multi-agent pipeline behind a human-approval gateway, Google OAuth + JWT, Stripe billing, rate limiting, and spend-cap / kill-switch safety controls.

bhasha-js | TypeScript, React, npm

npmjs.com/package/bhasha-js

- Published a drop-in internationalization (i18n) SDK to npm that intercepts website text at the component level and serves AI-generated translations, requiring zero per-language developer work.
- Designed a lazy-translation model: untranslated strings are translated on first request, cached for subsequent visitors, and auto-refresh when the source text changes.

SKILLS

- **Languages:** Python, Go, TypeScript, JavaScript, Java, SQL
- **Backend & Cloud:** Node.js, Express.js, Flask, FastAPI, PostgreSQL, MongoDB, Redis, AWS, Docker, CI/CD
- **Frontend:** React, Next.js, Tailwind CSS
- **ML/AI & Testing:** PyTorch, CLIP/YOLO, LLM APIs, Git, Linux, pytest, Playwright, Jest